

WASP-163b

R-2394

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-163_260703 · created 2026-08-20T23:54:16+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461224.774 +/- 0.074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.123 +/- 0.04
Transit depth (Rp/Rs)^2	1.51 +/- 0.99 [%]
Orbital Inclination (inc)	82.98 +/- 2.73
Airmass coefficient 1 (a1)	1.127 +/- 0.018
Airmass coefficient 2 (a2)	-0.074 +/- 0.027
Scatter in the residuals of the lightcurve fit is	1.61 %
Best Comparison Star	#2 - [610, 91]
Optimal Aperture	3.0
Optimal Annulus	7.12
Transit Duration (day)	0.039 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.123 ± 0.04 vs 0.119 (deviation 0.1σ)
Duration fit vs published	0.039 d vs 0.0942 d (deviation 1.67σ)
Mid-time O–C	16.3 min
Depth SNR	3.1
Residual scatter	1.61 %

Light curve

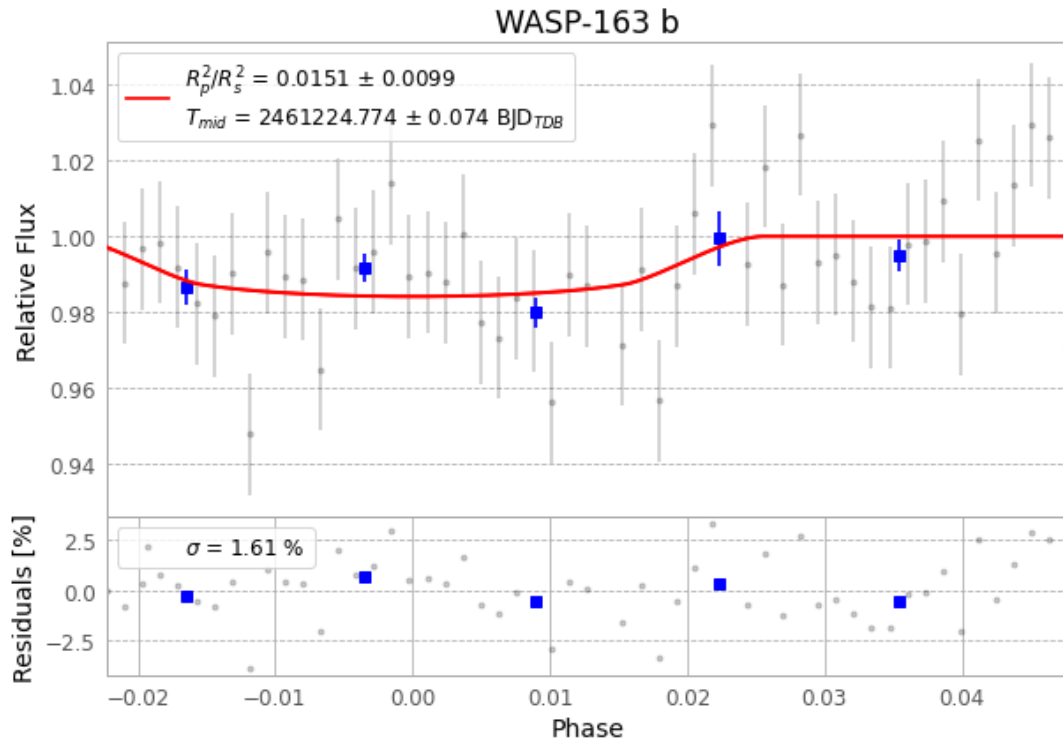


Figure 1 — FinalLightCurve_WASP-163 b_2026-07-02.png (SHA-256 b7afa7fcb8cf8fab...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [286, 136], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 95586218cd3ba402...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

55 FITS frames (36 MB), WASP-163260703053315.FITS → WASP-163260703081515.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-163-260703.log (ab24f4040014...), FinalLightCurve_WASP-163 b_2026-07-02.png (b7afa7fcb8cf...), FinalLightCurve_WASP-163 b_2026-07-02.csv (dce9a40c5735...)

Compute resources

Wall time 92.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-20 23:54Z.